WOW!

HDB PROPERTIES

# HDB Resale Price Prediction

## Using Machine Learning to Empower Singapore's HDB Resale Buyers and Sellers

*Image: The Pinnacle@Duxton, Singapore | Credit: Zairon, CC BY-SA 4.0, via Wikimedia Commons*

ANY Debs | GA Data Analytics Bootcamp | Project 3

# Executive Summary

*What we built, why it matters, and what WOW! gets*

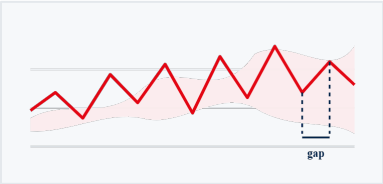
OUR APPROACH

## From problem to deployed tool — in four steps.

### STEP 01

THE PROBLEM

**Volatile market, broken pricing**




**$400K +**

LOCATION PRICE GAP ACROSS PLAN AREAS

### STEP 02

THE DATA

**167,000+ transactions, 77 features**

**9 years**

HDB RECORDS 2012–2021

### STEP 03

THE MODEL

**Random forest, 300 trees**

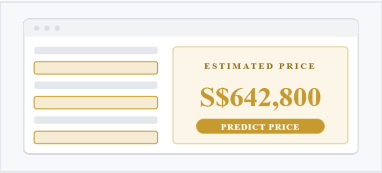
**90% +**

PREDICTION ACCURACY

### STEP 04

THE TOOL

**Live Streamlit Price Predictor**




**11 inputs**

INSTANT RESALE PRICE ESTIMATE – **DEPLOYED AND READY FOR WOW! AGENTS AND CLIENTS**

# Who is WOW!

*Singapore's HDB Specialists – a profile of the firm*

COMPANY PROFILE

## WOW! HDB Properties Division

*Image: Singapore's HDB Flats.jpg2 | Credit: Justbemystar, CC BY-SA 4.0, via Wikimedia Commons*

### About the company

Founded in **2008** in Tampines, **WOW! Properties** is a boutique Singapore real-estate agency with a singular focus: helping everyday Singaporeans transact HDB resale flats with confidence. Our dedicated **HDB Properties Division** handles 3-, 4-, 5- room and executive units across all 32 planning areas.

**17 years**
SERVING SINGAPORE FAMILIES

**42**
LICENSED AGENTS ISLAND-WIDE

**S$1.4B**
HDB VOLUME · 2025

CEA REGISTRATION
**L3009876K**

HEADQUARTERS
**Tampines Central, SG**

HDB DIVISION EST.
**2010 · 15 years**

CLIENTS SERVED
**5,300+ families**

# The ANY Debs Team

*The WOW! HDB Properties team at your service*

ANY DEBS TEAM

## Four agents. One sharper way to price HDB flats.

A decade of combined transaction experience across Singapore's HDB market. Every member of this team has closed at a level their peers benchmark against — and every one will use the price predictor on client engagements from day one.

**"Data-led listings close faster — and at fairer prices for both sides."**

| EXPERIENCE | CLOSED 2024 |
|---|---|
| **11 yrs** | **S$1.3M** |

SPECIALTY

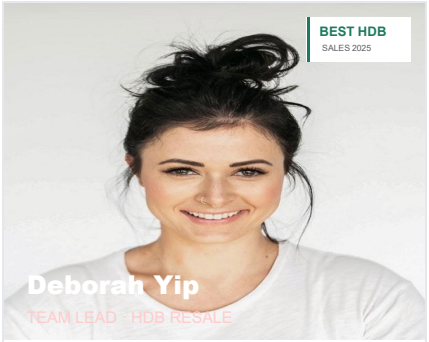
Queenstown · Bukit Merah · Central mature estates

**"Every family deserves a price they can defend with numbers, not opinions."**

| EXPERIENCE | CLOSED 2024 |
|---|---|
| **15 yrs** | **S$5.4M** |

SPECIALTY

Executive & maisonette flats · Tampines · Pasir Ris




**"Being named Best HDB performer was built on evidence — we bring the same rigour to your flat."**

| EXPERIENCE | UNITS 2024 |
|---|---|
| **9 yrs** | **38 sold** |

SPECIALTY

First-time upgraders · Punggol · Sengkang BTO resales

**"Young buyers want transparency. This model gives them exactly that — upfront."**

| EXPERIENCE | CLOSED 2024 |
|---|---|
| **7 yrs** | **S$1.1M** |

SPECIALTY

4-room resales · Hougang · Ang Mo Kio · Bishan

# Singapore's HDB Market

*The World's Most Successful Public Housing*

## Context & Stakes

- WOW! agents advise hundreds of clients per yeareven a 5% pricing error translates to S$25,000+ on a typical S$500K flat.
- For most Singaporean families, an HDB flat is their largest single asset — accurate pricing is high-stakes.
- HDB resale prices are perceived to be acutely sensitive to floor level, remaining lease, proximity to MRT, schools, malls and hawker centres.

*Image: Ghim Moh panorama, Singapore - 20101229.jpg | Credit: User:Chensiyuan, CC BY-SA 4.0, via Wikimedia Commons*

**~80%**
of Singaporeans live in HDB flats

**1M+**
HDB units across Singapore

**$500K**
Median resale price (2021)

# The Pricing Challenge

*Why intuition is not enough*

## The Agent's Dilemma

- Prices vary by S$200-400K between planning areas for the same flat size.
- A 5-room flat in Queenstown can cost 60% more than an identical unit in Woodlands.
- MRT proximity alone can swing prices by 57% — from S$290K (3.5km away) to S$457K (within 500m).
- HDB age (lease decay) creates non-linear price drops that agents struggle to quantify.
- Current agent tools rely on recent comparables — no systematic feature-weighting.

*Image: https://www.flickr.com/photos/erwin_soo/8497369921 | Credit: Erwin Soo from Singapore., CC BY 2.0 via Wikimedia Commons*

**The Opportunity:**

A predictive ML model trained on 167,000 historical transactions can systematically quantify every one of these factors — giving WOW! agents a competitive advantage over other property agents.

# Why WOW! Needs This Tool

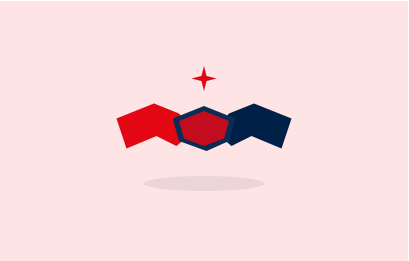
*Accuracy beyond intuition – the business case*

THE BUSINESS CASE

## Four compounding advantages, one shared tool.

### Competitive Edge

Price listings more accurately than competitors who rely on gut feel and comparable sales alone — winning more listings at the pitch.

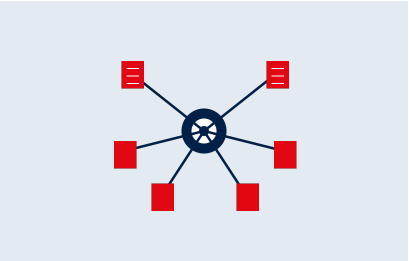


### Client Trust

Sellers and buyers trust agents who back pricing advice with data. Evidence-based conversations build long-term relationships and referrals.

### Faster Deals

Accurately priced listings sell faster. Reduced days-on-market directly improves agent throughput and commission velocity.



### Scalability

One model serves every WOW! agent across all 24 planning areas and every flat type — no extra training required for new estates.

# The Dataset

*150,000+ transactions, 2012 to 2021 — 10 years of Singapore's HDB resale market*

| 150,634 | 78 | 16,737 | 2012-21 | 9 |
|---|---|---|---|---|
| Training transactions | Feature columns | Validation transactions | Time span | Feature groups |

## Train / Validation Split

- train.csv: 150,634 rows x 78 columns. Includes resale_price (target).
- test.csv: 16,737 rows. — target variable (resale_price) is not available
- Key difference: train set has resale_price column not in test.
- Both datasets are aligned to the same feature structure using column reindexing (fill_value=0).
- Target variable: resale_price in SGD — ranges from ~S$150K to ~S$1.3M.

## 9 Feature Groups at a Glance

1. Flat Characteristics (flat_type, floor_area_sqft, storey_range, full_flat_type)
2. Location & Geography (town, planning_area, lat/lon, postal)
3. MRT & Bus Connectivity (mrt_nearest_distance, interchange flags)
4. Shopping Malls (Mall_Within_500m / 1km / 2km)
5. Hawker Centres (Hawker_Within_500m / 1km / 2km)
6. Primary Schools (pri nearest distance, Vacancy)
7. Secondary Schools (sec nearest distance, cutoff points)
8. Block Features (commercial, carpark, precinct_pavilion)
9. Temporal & Lease (Tranc_YearMonth, hdb_age, lease_commence)

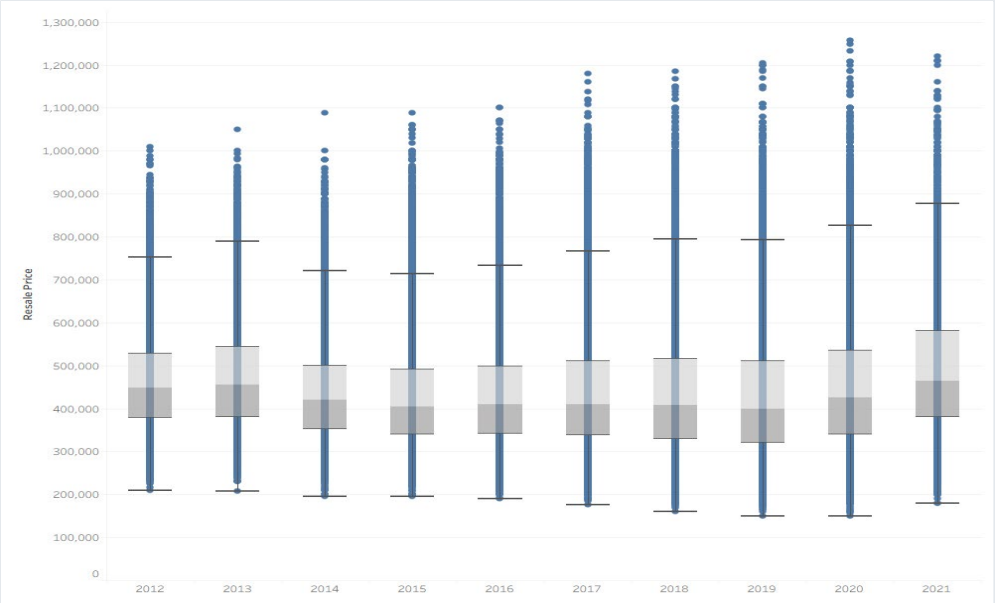
# Resale Price Trends

*10 years of HDB Resale Prices – three distinct market areas*

DISTRIBUTION · 2012 – 2021

## A decade in three acts: peak, plateau, pandemic surge.

RESALE PRICE BY YEAR · ALL HDB FLAT TYPES




2012 – 2013 · PEAK

### Relatively high prices

Median stable **above S$400K** . Upper whiskers climbed, suggesting the ceiling for expensive flats was rising.

2014 – 2019 · PLATEAU

### Flat period

IQR visibly lower — flats more affordable. Median level **around S$400K** . Outliers continued to creep upward.

2020 – 2021 · SURGE

### Pandemic upward shift

BTO construction delays and low global interest rates lifted borrowing power. Distribution shifts up — widening IQR.

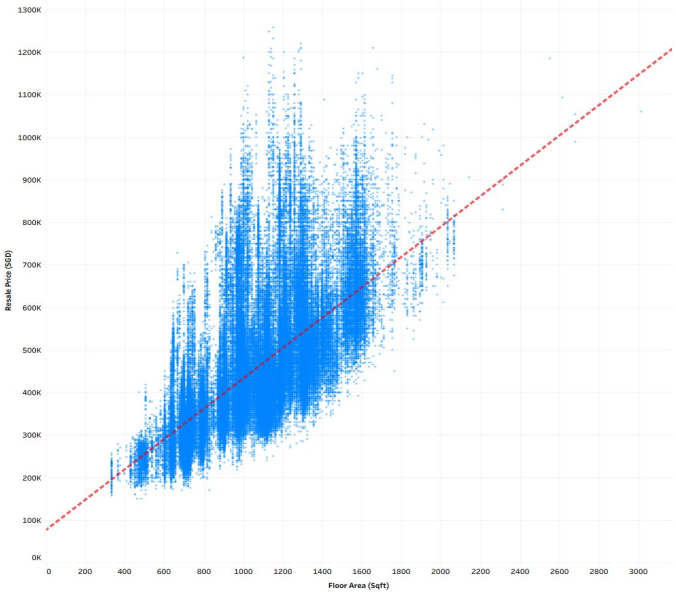
**New median peak: S$465K (2021)**

# Floor Area — The Foundation of HDB Pricing

## Key Findings

- Larger flats (Executive, 5-room) consistently command higher prices.
- $R^2 = 0.43$ means 57% of price variance is explained by other factors (location, age, flat model, proximity).
- The relationship is positive and roughly linear up to ~1,400 sqft, then flattens for very large executive units.
- Vertical scatter around the trend line reveals location and flat model effects.
- p-value < 0.0001 confirms the relationship is statistically robust, not chance.

## LARGER FLATS COMMAND BETTER RESALE PRICES



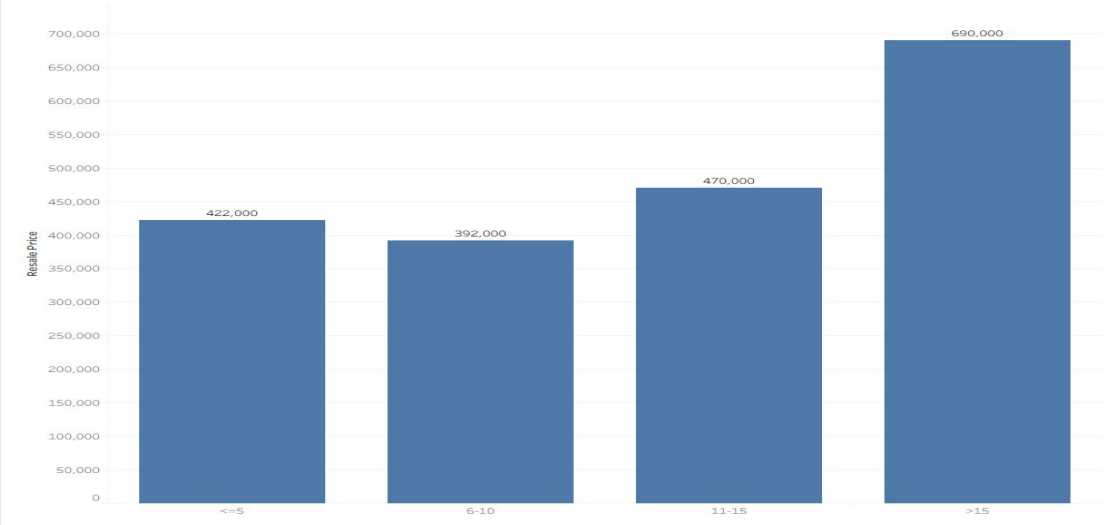

**$R^2 = 0.4279$ (~43%)**
There is a statistically significant positive relationship between floor area and resale price. Floor area explains ~43% of the variation in resale prices. Floor area is an important driver of price, but it is not sufficient on its own to explain pricing—other factors such as location, flat type, and amenities play a significant role.

**p-value < 0.0001**
This is extremely significant. The relationship between floor area and price is statistically robust.

# The Hawker Signal — 18.4% of the Model Runs on CKT

*Hawker_Within_2km is the #2 feature by importance — but it's not really about the food*

## MEDIAN RESALE PRICE BY HAWKER CENTRES WITHIN 2KM




### 0 – 10 Hawkers — S$392-422K

**SUBURBAN / OUTLYING**

Low hawker density = outlying estates. Prices reflect non-mature, lower-land-value zones.

### 11 – 15 Hawkers — S$470K

**MID-TIER RESIDENTIAL**

Moderate hawker density = established but not central estates. Prices reflect location maturity and accessibility.

### >15 Hawkers — S$690K

**MATURE CENTRAL — PREMIUM**

Mature central estates. Hawker-dense AND centrally located AND land-scarce — a triple premium.

**THE REAL STORY:** Hawker_Within_2km is a proxy for mature estate density — not a direct price driver. The model uses it as a powerful location signal because central estates happen to have both the most hawker centres and the highest land values. For WOW! agents: a flat near 15+ hawker centres is almost certainly in a mature central estate — price accordingly.

# Flat Model Matters — 3-Room Pricing Tiers

*Full flat type is more predictive than flat type alone*

## Prestige — S$810K

**Types: Terrace Houses**

These are extremely rare, landed HDB properties

## Modern & Premium — S$375K–500K

**Types: Premium Apartment, DBSS**

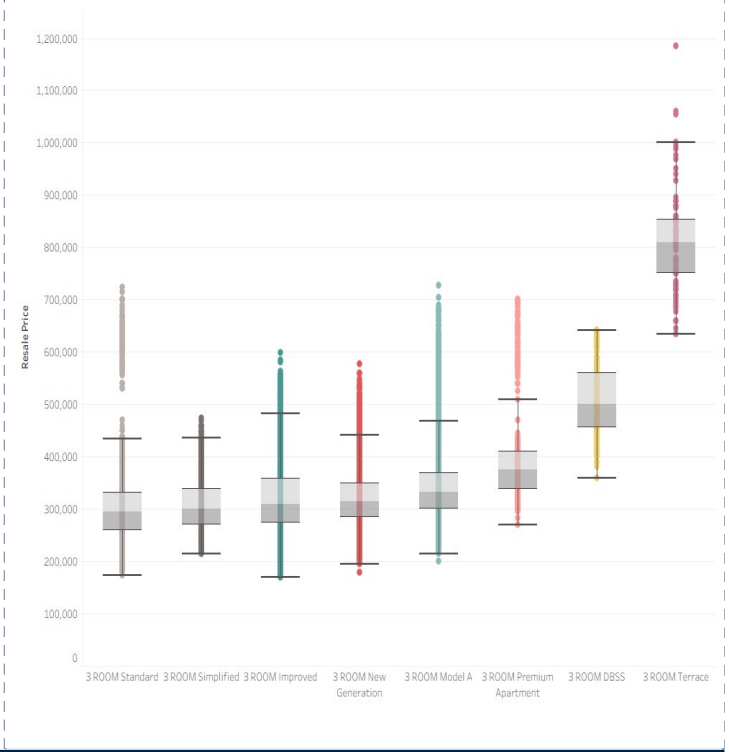
These units feature higher-quality finishes (like "condo-like" flooring or windows) and are often part of newer developments built by established developers.

## Standard & Functional — S$295K–333K

**Types: Standard, Simplified, Improved, New Generation, Model A**

Base HDB models with standard layouts. Pricing here driven almost entirely by location, floor level, and remaining lease rather than flat model.

## RESALE PRICE BY FULL FLAT TYPE




WHY THIS MATTERS: full_flat_type (40+ one-hot dummies) is one of the strongest feature groups in the Random Forest model — more predictive than flat_type alone because it captures model-specific premium effects.

# HDB Age & The 99-Year Lease Decay Effect

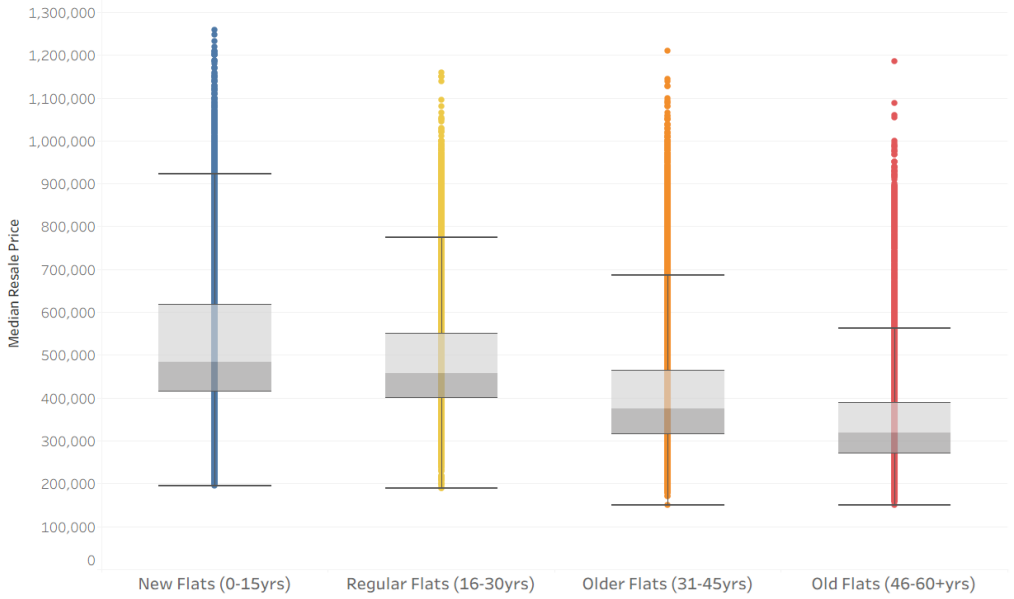
*Older flats lose value non-linearly — a critical signal for seller expectation management*

**KEY INSIGHT:** HDB flats operate on a 99-year leasehold. As remaining lease falls below 60 years (or older than 30 years), CPF usage restrictions and bank financing limits kick in — creating non-linear price drops that models must capture.

## Age Impact on Financing, Resale Price and Discounting

| | |
|---|---|
| **0–15 years** 84–99 yrs left | No restrictions. Full CPF & bank loan eligibility. |
| **16–30 years** 69–83 yrs left | Minor concern. Still fully financeable. |
| **31–45 years** 54–68 yrs left | Buyers begin to factor lease decay into offers. |
| **46–60 years** 39–53 yrs left | CPF restrictions apply. Bank loans shortened or unavailable, noticeable price discount. |

## OLD FLATS HAVE TO BE DISCOUNTED TO PRICE IN FINANCING HURDLES



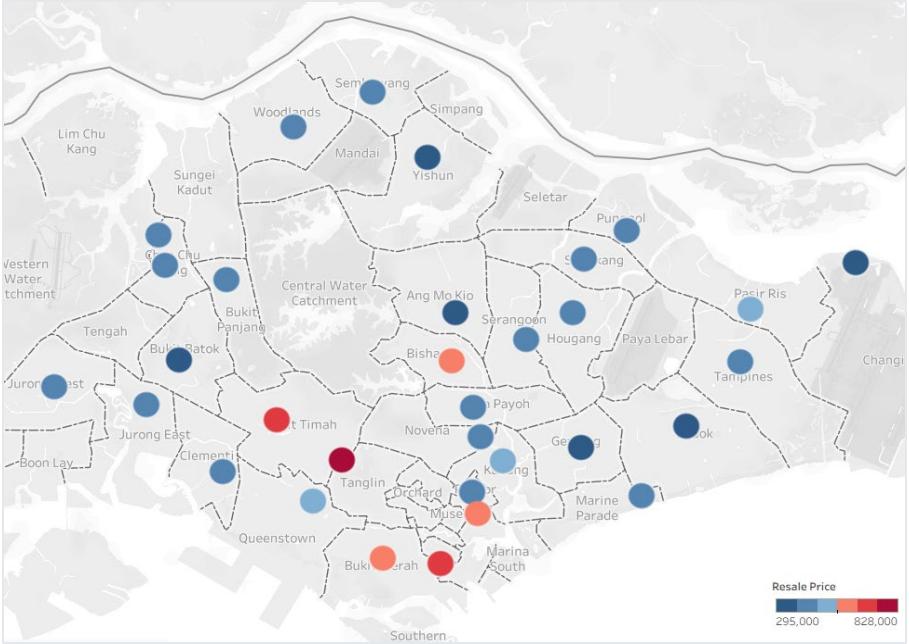
# Where Location Sets The Price

*Central and mature estates command S$200–400K premiums over northern and western towns*

GEOGRAPHY · PRICE PREMIUM

## Postcodes shouldn't cost as much as properties — but in Singapore, they do.

MEDIAN RESALE PRICE · BY PLANNING AREA




### Top planning areas by median price

Central location, maturity, and land scarcity drive prices to the top of the scale.

| PLANNING AREA | CLASSIFICATION | MEDIAN |
|---|---|---|
| Tanglin | MATURE · CENTRAL | S$828K |
| Bukit Timah | MATURE · CENTRAL | S$706K |
| Outram | MATURE · CENTRAL | S$680K |
| Bishan | MATURE · NEAR-CENTRAL | S$595K |
| Bukit Merah | MATURE · CENTRAL | S$568K |
| Woodlands | NON-MATURE · NORTH | S$380K |
| Jurong West | NON-MATURE · WEST | S$370K |

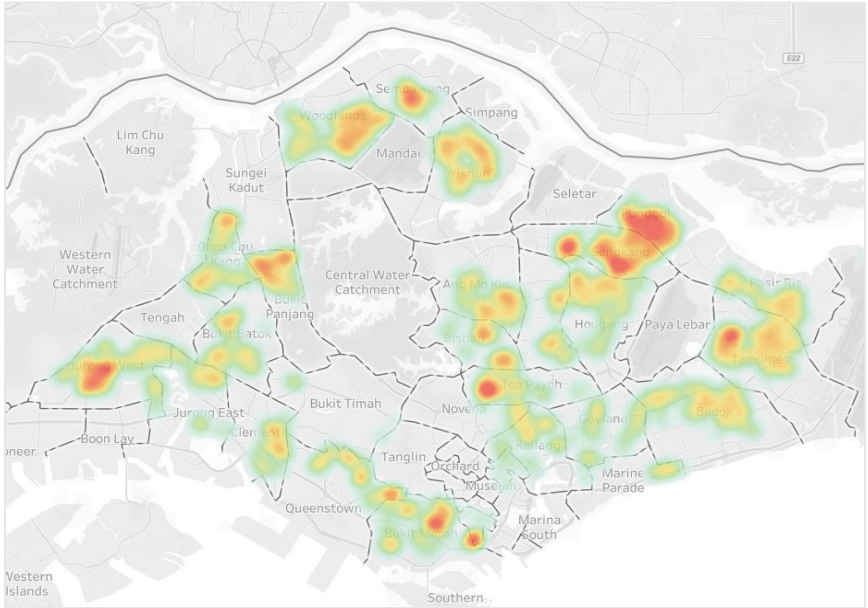
# Where Singapore Trades HDB Flats

*The busiest estates aren't the priciest — volume and premium tell different stories*

LIQUIDITY · TRANSACTION DENSITY

## High-volume estates signal liquidity. Mature estates signal premium.

TRANSACTION DENSITY HEATMAP · 2012–2021




HIGH DENSITY · MEDIUM · LOW

HOTSPOTS: SENGKANG · PUNGGOL · TAMPINES · JURONG WEST · WOODLANDS

### Volume ≠ Price

Highest-volume estates — **Sengkang, Tampines, Punggol** — are not the priciest. Volume signals liquidity; ease of exit from the asset, not its market value.

### Mature estate premium

Established estates — **Queenstown, Toa Payoh, Bishan** — show lower transaction volume but command significantly higher per-unit prices driven by scarcity and central access.

### Agent strategy — two playbooks

In **high-volume estates**, emphasise liquidity and school access. In **mature estates** , emphasise asset appreciation history.

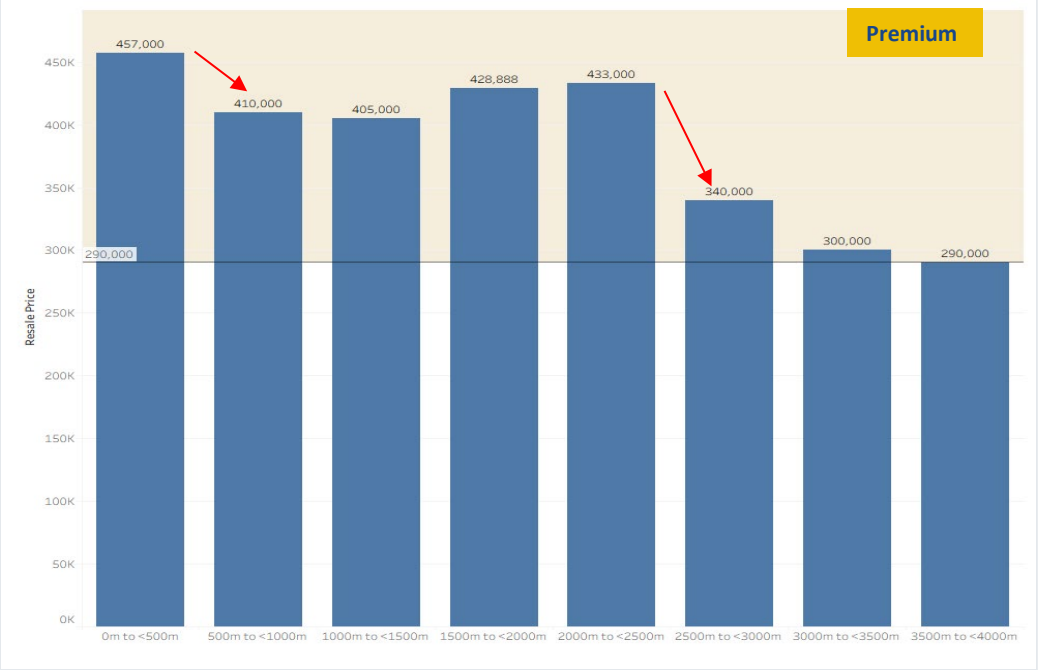
# The MRT Proximity Premium

*Distance to the nearest MRT explains a large share of price variation*

CONNECTIVITY · MRT DISTANCE

## Extreme proximity drives premium pricing — a 5-minute walk is worth S$167K.

MEDIAN RESALE PRICE BY DISTANCE TO MRT · METRES




**S$457K**

< 500M · PREMIUM

**Within a 5–7 min walk**

Buyers pay a heavy premium for extreme proximity. Median price is the highest across all tiers.

**–10%**

500M – 2.5KM · SOFT DROP

**Walkable, not premium**

Price drops ~10% once distance from MRT moves past 500m, then stabilises across the mid-range.

**–21%**

> 2.5KM · PENALTY

**Commute becomes unattractive**

Price plummets 21% beyond 2.5km. Buyers discount heavily once the flat is no longer transit-friendly.

# Data Pipeline

*From raw CSV to model-ready features in 4 stages*

PIPELINE · END TO END

## Four transformation stages. Identical on train and test.

**STAGE 01**
### Raw Data
- train.csv (150,634 rows)
- test.csv (16,737 rows)
- 77 columns each

›

**STAGE 02**
### Clean
- Type conversion (boolean, datetime, numeric, string)
- Null audit + zero-fill proximity
- Deduplication

›

**STAGE 03**
### Feature Engineer
- Proximity boolean flags
- Count boolean flags
- One-hot encoding (75 dummies)

›

**STAGE 04**
### Model Ready
- X_train, y_train
- X_test, y_test
- ~81 feature columns

### No Data Leakage

clean_data() and engineer_features() applied to BOTH train and test independently.
No statistics from test used to transform train.

### Identical Schema

df_test.reindex(columns=df_train.columns, fill_value=0) ensures column order and presence is 100% identical for model.predict().

### Reproducible

All transformations wrapped in functions. Random state set to 42 for train_test_split.
Pipeline is fully reproducible end-to-end.

# Feature Engineering

*Three transformations turn raw data into model-ready signals*

## 1. Proximity Boolean Flags

For each distance variable (mrt_nearest_distance, pri_sch_nearest_distance, sec_sch_nearest_dist), four threshold flags created:

_500m_bool | _1km_bool | _2km_bool | _2kmPlus_bool

Captures threshold effects: being within 500m of MRT = meaningful premium.

**EXAMPLE**

```
mrt_nearest_distance = 420m
→ _500m_bool = 1
→ _1km_bool = 1
→ _2km_bool = 1
→ _2kmPlus_bool = 0
```

## 2. Count Boolean Flags

For Mall_Within_Xkm and Hawker_Within_Xkm count columns, binary flags (>0) created:

Mall_Within_500m_bool
Mall_Within_1km_bool
Mall_Within_2km_bool
Hawker equivalents (x3)

Converts sparse count distributions into clean presence/absence signals.

**EXAMPLE**

```
Mall_Within_1km = 3
→ Mall_Within_1km_bool = 1
Mall_Within_500m = 0
→ Mall_Within_500m_bool = 0
```

## 3. One-Hot Encoding

pd.get_dummies() applied to:

full_flat_type (type + model): 43 dummies
planning_area (URA zone): 32 dummies

Total encoded features: 75 additional binary columns.

Train/test aligned via: df_test.reindex(columns=df_train.columns, fill_value=0)

**EXAMPLE**

```
4 ROOM Model A
→ full_flat_type_4 ROOM Model A = 1
→ all other type dummies = 0

Bishan planning area
→ planning_area_Bishan = 1
```

# Model Selection – Linear Regression Vs Randon Forest

*Comparing Linear Regression (fast/interpretable) vs. Random Forest (accurate/robust)*

## Model 1 — Linear Regression

- Goal: Minimize Ordinary Least Squares
- Scaling: StandardScaler in Pipeline (required for coefficient interpretability)
- Interpretation: Coefficients show feature direction + magnitude
- Strength: Extremely fast to train; highly interpretable; good baseline
- Limitation: Assumes linear relationships; cannot capture flat model x location interaction effects
- CV $R^2$: ≈0.86 (strong but outperformed by RF)

## Model 2 — Random Forest

- Goal: Individual trees tuned to answer its own sample of the data set with high accuracy followed by reducing variance by averaging results
- Hyperparameters: 300 trees, 12 decision splits, has 10 samples before each spilt, has at least 4 samples after every split
- Interpretation: feature_importances of what each tree used to train itself to be more accurate
- Utilizes Non-Linear Relationships: Captures non-linear interactions (e.g. MRT proximity x planning area); robust to outliers
- Averaged weights : No scaling required; handles mixed numeric and binary features natively
- CV $R^2$: >0.9 on test set — selected as production model

**DECISION: Random Forest selected —superior accuracy ($R^2$>0.9), captures non -linear feature interactions, and provides feature importance rankings directly usable for agent training.**

# Evaluation & Head-to-Head Results

*How we measured accuracy — and why Random Forest won by 40%+ on MAE*

VALIDATION · HEAD TO HEAD

## Random Forest beats Linear Regression on every metric.

### How we measure

**CV $R^2$ (10-FOLD)**
**Mean $R^2$ across 10 folds**
Price variance explained on unseen data. Trains on 9 minisets, tests on the 10th, repeated.

**VALIDATION $R^2$**
**Held-out test $R^2$**
Final out-of-sample accuracy. Confirms generalisation.

**MAE · RMSE**
**Mean / Root-Mean-Sq Error**
MAE: avg dollar error. RMSE: penalises large errors more.

| METRIC | LINEAR REGRESSION | RANDOM FOREST [SELECTED] |
|---|---|---|
| CV $R^2$ (mean ± std) | 0.8654 (± 0.0016) | 0.9085 (± 0.0020) |
| Validation $R^2$ | 0.8635 | 0.9097 |
| Validation MAE (S$) | 41,269 | 33,382 |
| Validation RMSE (S$) | 52,916 | 43,045 |
| Train MAE (S$) | 40,995 | 32,834 |

**WHY RANDOM FOREST WINS**

**Non-linear interactions** (Queenstown + large flat = outlier premium).

**Robustness** (outliers smoothed by 300-tree averaging).

**Feature importance** (actionable rankings for agent training).

# Accuracy Beyond Intuition – Actual vs. Predicted Prices

*300 trees produce predictions that closely mirror market reality*

MODEL VALIDATION · SCATTER

## Predictions hug the 45° line — the model sees what the market sees.

RANDOM FOREST · ACTUAL VS PREDICTED HDB RESALE PRICES

HOW TO READ THE CHART

### Hug the line = win

Each dot is a held-out transaction. The **dashed red line** is a perfect prediction. Tight clustering around the line means the model matches reality across the full S$200K–1.2M price range.

**ABOVE the line · Overprediction**

Model priced the flat **higher than it sold**. Risk for **sellers** — listing may sit on the market.

**BELOW the line · Underprediction**

Model priced the flat **lower than it sold**. Risk for **buyers** — we adjust upward with recent comps.

TAKEAWAY

Dispersion widens slightly above S$900K where sample size thins — agents should pair model output with recent premium-tier comps for flats over S$1M.

# How Random Forest Works

*300 trees vote together to produce a price estimate no single agent could compute*

THE ALGORITHM · FOUR STEPS

## Bootstrap → split → branch → average.

### 01

**Bootstrap Sampling**

Each tree trains on a random bootstrap sample from 150,634 training rows. ~63.2% unique rows per tree; ~36.8% become Out-of-Bag (OOB) for validation.

### 02

**Random Feature Selection**

At each split, only a random subset of the ~81 features is considered. Each split picks the optimal feature + threshold that best reduce Mean Square Errors.

### 03

**Branching**

Branching continues until stopping criteria — max_depth, min_samples_split, min_samples_leaf.

### 04

**Averaging**

All 300 trees predict independently. The final prediction is the **mean** of all 300 estimates — averaging out "biased" individual tree noise to generalise.

**HYPERPARAMETERS & RATIONALE**

| n_estimators | max_depth | min_samples_split | min_samples_leaf | random_state | cross-val |
|---|---|---|---|---|---|
| **300** | **12** | **10** | **4** | **42** | **10-fold** |
| Strong variance reduction without over-training | Prevents overfitting on training noise | Prevents trees fitting outliers | Stabilises predictions in sparse regions | Full reproducibility across runs | Stable $R^2$ estimate: mean ± std |

# What The Model Learned – 8 Top Feature Importances

*The features that matter most — and the business insights they unlock*

| | | |
|---|---|---|
| #1 | `floor_area_sqft` 53.1% | Largest individual driver. Every sqft matters — agents must quantify floor area precisely. |
| #2 | `Hawker_Within_2km` ~18.44% combined | Uniquely Singaporean. Hawker access is a genuine lifestyle signal the model picks up. |
| #3 | `full_flat_type_* dummies` ~8.49% combined | Type + model explains more than type alone. DBSS and Premium dummies carry the highest weights. |
| #4 | `hdb_age` ~7.9% | Lease decay effect. Model has learned the non-linear relationship between age and value. |
| #5 | `planning_area_* dummies` ~6.56% | Location remains as an important factor. The model 'knows' each of Singapore's 32 planning zones. |
| #6 | `mrt_nearest_distance` ~2.68% | The 500m rule shows up in feature importance. Proximity to MRT is objectively quantified. |
| #7 | `mid_storey` ~1.98% | Higher floors = higher prices. Each additional storey adds measurable value. |
| #8 | `Mall_Within_2km` ~0.79% | Urban amenity signal. Buyers in mature estates value retail access strongly. |

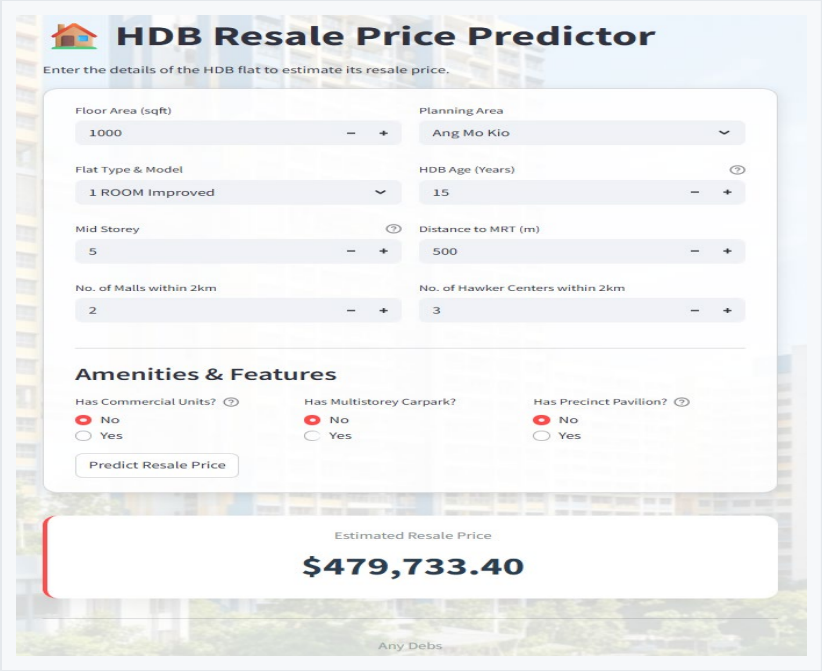
# HDB Resale Price Predictor

*A Streamlit tool agents can use at the listing conversation*

THE TOOL · FOUR FEATURE FAMILIES

## Enter twelve attributes. Get an evidence-based price range in seconds.

STREAMLIT UI · LIVE EXAMPLE · ANG MO KIO




### Geographical & Physical

Where the flat sits — planning area & floor level — and how much of it there is.

- PLANNING AREA
- MID STOREY

### Unit Specifications

Intrinsic flat attributes: size, model, and age of the lease.

- FLOOR AREA (SQFT)
- FLAT TYPE & MODEL
- HDB AGE

### Connectivity

How well-linked the unit is to daily transit — the single biggest price driver.

- DISTANCE TO MRT (M)

### Lifestyle Premium

Neighbourhood amenities and on-estate perks that add measurable premium.

- MALLS WITHIN 2KM
- HAWKER CENTRES WITHIN 2KM
- COMMERCIAL
- MULTISTOREY CARPARK
- PAVILION

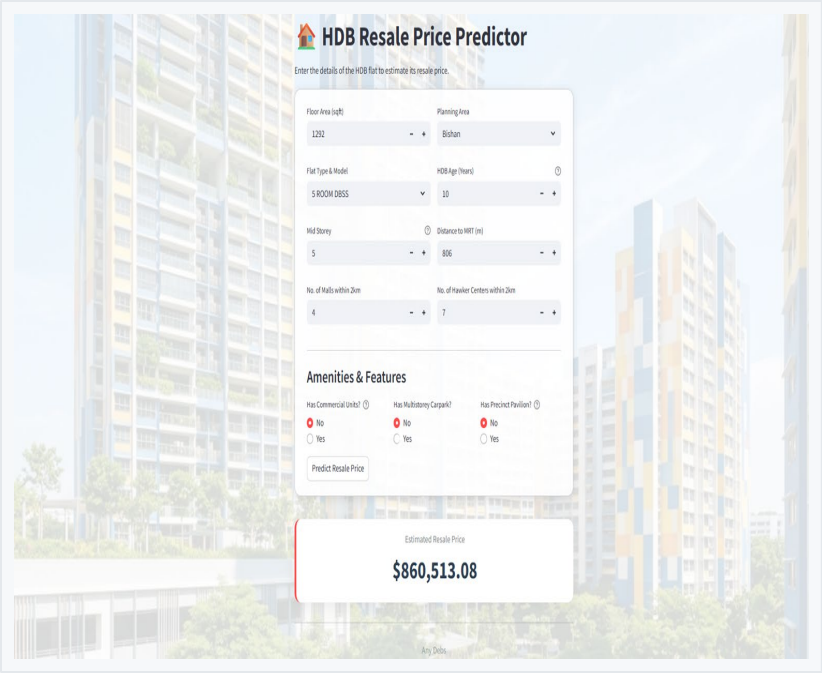
# Test Data Results

*A held-out Bishan 5-room DBSS: model prediction vs. actual*

VALIDATION · HELD-OUT SAMPLE

## Bishan · 5-ROOM DBSS · 1292 sqft — predicted within the expected band.

LIVE PREDICTION · STREAMLIT TOOL




ACTUAL RESALE PRICE

**S$919,999.08**

Held-out Bishan 5-ROOM DBSS transaction — not seen during model training.

MODEL EXPECTED PRICE RANGE

**S$827,130.83 → S$893,895.33**

| POINT PREDICTION | VARIANCE VS. ACTUAL | VARIANCE VS. UPPER BAND |
|---|---|---|
| S$860,513 | 6.9% | 2.9% |

| VALIDATION MAE | VERDICT |
|---|---|
| S$33,382.25 | Within expected band — model validated |

# Business Impact – What WOW! Gains

*The agent workflow before vs after the Streamlit Price Predictor*

IMPACT · BEFORE VS AFTER

## One tool. Six compounding gains for every listing.

BEFORE · INTUITION & COMPS

### The old workflow

| | |
|---|---|
| PRICING METHOD | **Gut + 3–5 comps**<br>Agents anchored on recent nearby transactions. No systematic weighting of flat model, lease, or connectivity. |
| PREP TIME | **2–3 hours / listing**<br>Manually pulling comparable sales, adjusting by feel. |
| PRICING ERROR | **~10–15% typical**<br>S$50K–75K on a S$500K flat. Listings sit or under-sell. |
| CLIENT TRUST | **"Trust me, I've done this 10 years"**<br>Hard to defend pricing when challenged. Referrals on reputation alone. |
| DAYS ON MARKET | **45–60 days avg**<br>Mispriced listings attract low offers or no viewings. |
| SCALABILITY | **Tacit, per-agent**<br>Each agent's estate knowledge is siloed. New hires ramp slowly. |

AFTER · STREAMLIT + RF MODEL

### The new workflow

| | |
|---|---|
| PRICING METHOD | **Model + 167K transactions**<br>**90% accuracy.** 81 features weighted systematically. |
| PREP TIME | **~2 minutes / listing**<br>**~60× faster.** Instant evidence-based range. More listings per agent per week. |
| PRICING ERROR | **MAE S$33K (~6%)**<br>**40%+ improvement on LR.** Predictions within the expected band. |
| CLIENT TRUST | **"Here's the evidence"**<br>**Defensible conversations.** Show the range, the features driving it, history. |
| DAYS ON MARKET | **Target 20–30 days**<br>**Faster deals → higher commission velocity.** Accurate prices attract serious buyers on day one. |
| SCALABILITY | **One tool · 42 agents · All Planning Areas**<br>**Institutional knowledge.** New hires are productive from day one; model retrains every 6 months. |

# Strategy & Recommendations

*Playbook for WOW! agents, powered by this model*

TURN DATA INTO DEALS

## Six plays. Every listing. Every agent.

### 01

**Lead with Floor Area & Storey**

Always anchor pricing on **floor area (sqft)** and storey level first — the two strongest individual predictors. Establish the 'base price' before layering location effects.

### 03

**Sell the Flat MODEL, Not Just the Type**

A 5-Room DBSS and a 5-Room Standard are priced very differently.
Train agents on the premium hierarchy: **DBSS > Premium Apt > Model A > Standard.**

### 05

**Location-Based Pricing Tiers**

Provide agents with a planning area pricing map. A 4-Room in Queenstown vs. Woodlands is a S$300–400K conversation — agents who can articulate this own the room.

### 02

**Quantify the 500m Rule**

For every listing, calculate and state the exact distance to the nearest MRT.
**"This flat is 420m from Tampines MRT — within the S$167K premium zone"** is a data-backed selling point every agent can use.

### 04

**Manage Lease Expectations**

For flats over 40 years old, proactively model financing constraints (CPF limits, shortened bank loan tenors) — set realistic expectations with sellers **before** listing.

### 06

**Deploy, Measure, Retrain**

Use the Streamlit tool in every client meeting for live estimates . Log actual transaction prices vs predictions quarterly. Retrain the model every 6 months to capture market shifts like the 2020–21 pandemic surge.

# Priced right. Sold fast.

Thank you. Questions welcome — and we'd love to run a live prediction on any flat you have in mind.

**Annie**
SENIOR ASSOCIATE DIRECTOR

**Newton Yap**
PRINCIPAL AGENT

**Deborah Yip**
TEAM LEAD · HDB RESALE

**Deb Sen**
SENIOR SALES CONSULTANT

# Model Limitations & Considerations

*Important constraints and usage guidelines for the HDB price prediction model*

## 01 Data Limitations

1. Historical data (2012–2021)
→ May not reflect current market conditions
2. Limited data for new developments
→ Reduced predictive accuracy for newer properties

## 02 Model Constraints

1. Limited Extrapolation Ability
→ Cannot predict beyond observed price range
→ Underestimates high-value properties
2. Dependence on Feature Engineering
→ Performance relies heavily on feature design

## 03 Geographic Coverage

1. Limited to HDB Resale Market
→ Excludes private properties
→ Model cannot generalize to full housing market
2. Uneven Area Representation
→ Some regions have fewer transactions
→ Predictions may be less reliable in these areas

## 04 Time Sensitivity

1. The housing market is dynamic
→ model performance may degrade over time
2. Seasonal and policy-driven variations partially captured

## 05 Usage Restrictions

1. For estimation purposes only
2. Not for legal/contractual decisions
3. Requires professional validation
4. Designed for WOW! agent support, not standalone valuation

## 06 Accuracy Metrics

1. Model: Random Forest (300 trees)
Validation $R^2$: > 0.90
2. Performance is strong within the observed data range

# Project Retrospective — Key Insights

*What we built, how we worked, and what we'd do differently*

SPRINT RETROSPECTIVE

## Three weeks. One shipped product.

- Wk 1: Data exploration, cleaning, EDA
- Wk 2: Feature engineering, model training
- Wk 3: Evaluation, Streamlit app, presentation

### WELL — What Went Well

- Modular code (clean_data, engineer_features, evaluate_model) made iteration fast and reproducible.
- Random Forest significantly outperformed Linear Regression on all metrics.
- Streamlit app deployed successfully with all 11 inputs and live predictions.
- Tableau dashboards effectively surfaced key pricing narratives for non-technical stakeholders.

### HARD — What Was Challenging

- Test set missing resale_price — reconstruction from price_per_sqft required careful validation.
- High dimensionality (~81 feature columns after OHE) slowed Random Forest training on local hardware.
- Ensuring identical column order between training pipeline and Streamlit app input vector.
- Trello sprint board management: proved difficult.
- Balancing EDA depth vs. modelling time vs visualization within the 3-weeks sprint constraint.

### LEARN — What I Learned

- Reusable evaluation functions are worth the upfront investment — every model run is just one function call.
- Feature importance from Random Forest is a powerful business communication tool, not just a metric.
- Agile sprint planning forces prioritisation — not everything in the product backlog gets done.
- Always align test schema to train after get_dummies().